

Kumar Abhishek

+91 96080 13812 abhishekkumar.ch2607@gmail.com LinkedIn GitHub Portfolio

PROFESSIONAL SUMMARY

Aspiring **Agentic AI / GenAI Engineer** with hands-on experience building **multi-agent systems**, **Model Context Protocol (MCP) servers**, **RAG pipelines**, and production-grade **LLM-powered applications**. Skilled in **LangGraph** agent orchestration, **tool calling**, **FastAPI**, **PyTorch/Transformers**, vector databases, and conversational AI systems that integrate external tools and APIs. Experienced in designing end-to-end AI workflows spanning retrieval, reasoning, autonomous tool execution, and deployment, backed by a solid foundation in ML model development and evaluation.

EDUCATION

Indian Institute of Information Technology, Una <i>B.Tech, Electronics & Communication Engineering</i>	Jul 2023 – Jul 2027 CGPA: 7.74/10
Delhi Model Public School, Delhi <i>Class XII (CBSE)</i>	Jun 2022 Score: 85%

PROJECT EXPERIENCE

AI Newsletter Automation – Multi-Agent MCP Server <i>Python FastMCP Model Context Protocol (MCP) Google APIs GitHub API</i>	Sept 2025 – Nov 2025 Github
- Built a production-ready Model Context Protocol (MCP) server using FastMCP, exposing 10+ callable tools that enable AI assistants to autonomously perform research, content generation, validation, and newsletter publishing through standardized tool-calling interfaces. - Designed a modular agent orchestration workflow coordinating Research, Editing, and Publishing phases, integrating arXiv, GitHub, Gmail, Google Drive, Product Hunt, and Twitter APIs for automated multi-source information retrieval and content curation. - Implemented reusable agent skills for paper retrieval, repository discovery, newsletter drafting, content organization, HTML generation, validation, cloud storage, and multi-format export, enabling end-to-end autonomous newsletter creation. - Deployed the solution as a hosted FastMCP Cloud server compatible with Claude Desktop, enabling zero-install integration through MCP while supporting scalable, production-ready AI workflows.	
InterviewBot – Agentic AI Interview Platform <i>FastAPI LangGraph LangChain Google Gemini ChromaDB MongoDB</i>	May 2025 – Nov 2025 Github Live
- Built an agentic AI interview platform using LangGraph to orchestrate stateful workflows across RAG-based chat, interview setup, and contextual assistance through multi-step agent execution. - Implemented RAG using ChromaDB and Sentence Transformers (all-MiniLM-L6-v2), alongside tool-calling pipelines for resume parsing, semantic retrieval, and personalized interview question generation from PDF/DOCX resumes. - Developed a JWT-secured FastAPI backend with asynchronous MongoDB integration, voice-enabled interactions, MCQ assessments, and automated PDF performance reports for personalized interview evaluation. - Integrated LangSmith for workflow tracing, prompt evaluation, debugging, and performance analysis of LangGraph agent execution, improving observability and iterative development.	
MediCure – AI-Powered Healthcare Assistant <i>FastAPI PyTorch Transformers Google Gemini React Tailwind CSS</i>	Jan 2025 – Jun 2025 Github Live
- Built a full-stack AI healthcare assistant combining fine-tuned BERT models and Google Gemini to provide medicine usage prediction, side-effect analysis, substitute recommendations, and conversational healthcare support. - Developed and deployed three PyTorch Transformer-based classification models for medicine usage (583 classes), side effects (1,271 classes), and substitute prediction (43,297 classes), exposed through a FastAPI backend for real-time inference. - Designed an AI-powered healthcare chatbot using Google Gemini 2.5 Flash with contextual symptom analysis, disease suggestions, diet recommendations, workout guidance, and precautionary advice. - Built a responsive React and Tailwind CSS frontend with REST API integration, delivering low-latency predictions across medicine analysis, home remedies, and conversational AI modules.	

TECHNICAL SKILLS

- Programming:** Python, SQL, Matlab.
- AI / ML / GenAI:** Machine Learning, Deep Learning, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Systems, Tool Calling, LangChain, LangGraph, LangSmith, MCP (Model Context Protocol), Transformers, Vector Databases, Agent-to-Agent (A2A) Protocol .
- Data Analysis & BI:** Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing, Data Cleaning, KPI Development, Dashboard Design, Power BI, Excel (Advanced).
- Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, Transformers, FastAPI, Streamlit, SciPy, ReportLab.
- Backend & Databases:** FastAPI, FastMCP, MongoDB, PostgreSQL, ChromaDB
- Tools & Platforms:** GitHub, Jupyter Notebook, Vercel, Hugging Face, Docker, AWS (Basic: EC2, S3, IAM).
- Coursework:** Machine Learning, Probability & Statistics, Linear Algebra.

POSITION OF RESPONSIBILITY

Active Member , AAVESH Technical Club, IIIT Una	2023 – Present
Active Member , EUNOIA Literature Club, IIIT Una	2023 – Present